CHAPTER-I

SUZON-50

F-2022

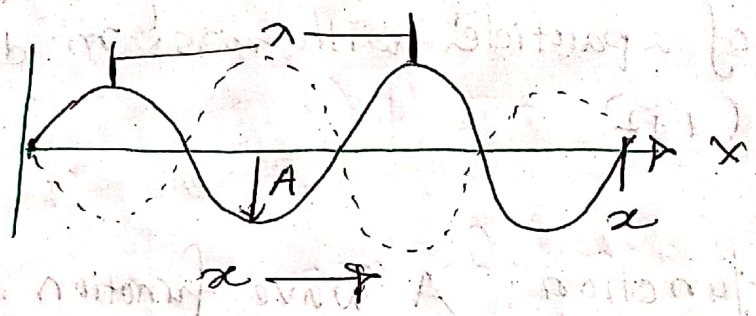
(a) What is a wave function? Deduce the general expression for Schrodinger wave equation for a standing wave motion of a particle with mass m defining each term in it. (1+2)

Ans:

Wave function: A wave function is a mathematical function that describes ~~that~~ the state of an electron within an atom or molecule, essentially representing the probability of finding an e- at a specific location in space and is denoted by ψ.

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Schrodinger wave equation : F-16



For a standing wave, in one dimension, the wave equation is given as —

$$\psi = A \sin 2\pi \frac{x}{\lambda} \quad \text{---------- (1)}$$

where, ψ = Wave function

λ = wave length

A = maximum amplitude

x = Space - coordinate / displacement of wave

$$\frac{d\psi}{dx} = \left(A \cos \frac{2\pi x}{\lambda}\right)\frac{2\pi}{\lambda}$$

$$= \frac{2\pi A}{\lambda} \cos \frac{2\pi x}{\lambda}$$

$$\frac{d^2\psi}{dx^2} = \frac{2\pi A}{\lambda} \cdot \frac{2\pi}{\lambda}\left(-\sin \frac{2\pi x}{\lambda}\right)$$

$$\Rightarrow \frac{d^2\psi}{dx^2} = -\frac{4\pi^2}{\lambda^2}\left(A \sin \frac{2\pi x}{\lambda}\right)$$

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So, $\frac{d^2\psi}{dx^2} = -\frac{4\pi^2}{\lambda^2}\psi$

$\Rightarrow \frac{d^2\psi}{dx^2} + \frac{4\pi^2}{\lambda^2}\psi = 0$ ..... (3)

According to De-Broglie, for a particle of mass m and velocity, v,

$$\lambda = \frac{h}{mv}$$

$$\Rightarrow mv = \frac{h}{\lambda} \quad \text{..... (4)}$$

$$\Rightarrow v = \frac{h}{m\lambda}$$

$$\Rightarrow v^2 = \frac{h^2}{m^2\lambda^2}$$

Again, we have, kinetic energy T is given as-

$$T = \frac{1}{2}mv^2$$

$$= \frac{1}{2}m \cdot \frac{h^2}{m^2\lambda^2} = \frac{1}{2}\frac{h^2}{m\lambda^2} \quad \text{..... (5)}$$

$$\Rightarrow \frac{1}{\lambda^2} = \frac{2m}{h^2} \cdot T \quad \text{..... (6)}$$

Total energy of a system, $E = T + V$ [V = potential energy]

$$\Rightarrow T = E - V \quad \text{..... (7)}$$

Now, (6) $\Rightarrow$

$$\frac{1}{\lambda^2} = \frac{2m}{h^2}(E - V) \quad \text{..... (8)}$$

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Putting the value of p_x in eqn (3), we get,

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2 \cdot 2m}{h^2}(E-V)\psi = 0$$

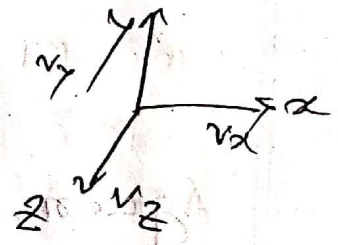
$$\Rightarrow \frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

This is the wave eqn for one dimension

Now for three dimensional case,

The relation between r and its component

$$r^2 = x^2 + y^2 + z^2$$



Now, the kinetic energy of the particle,

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2)$$

$$= \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$$

The schrodinger equation for this particle would be —

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

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It is also represented as

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

= is called Laplacian operator.

(b) What are Radial and Angular wave functions? Provide a graphical comparison between Radial wave function (R) and Radial Probability function (r^2R^2) for 1s orbital of H. (2+2)

Soln:

Radial Wave function:

The radial wave function determines the spatial extent of the orbital. It is denoted by $R_{n,l}(r)$. The radial wave function varies with r in a different way for each set of quantum numbers n and l.

Angular Wave function:

The wave function which determines the shape of the electron cloud and varies depending upon the type of orbital involved (s, p, d, f) and its orientation in space is called angular wave function.

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$$\psi_{l,m}(\theta,\varphi) = \Theta_{l,m}(\theta) \cdot \varphi_m(\varphi)$$

↓

Angular wave function.

Significance of Angular Wave function:

(I) The shape and orientation of an atomic orbit are largely determined by the angular wave function.

(II) The value of angular wave function in any direc is proportional to the length of a straight line extending from the origin up to the surf in the same direction

(III) It is also used to obtain information about the probable distribution of e^-.

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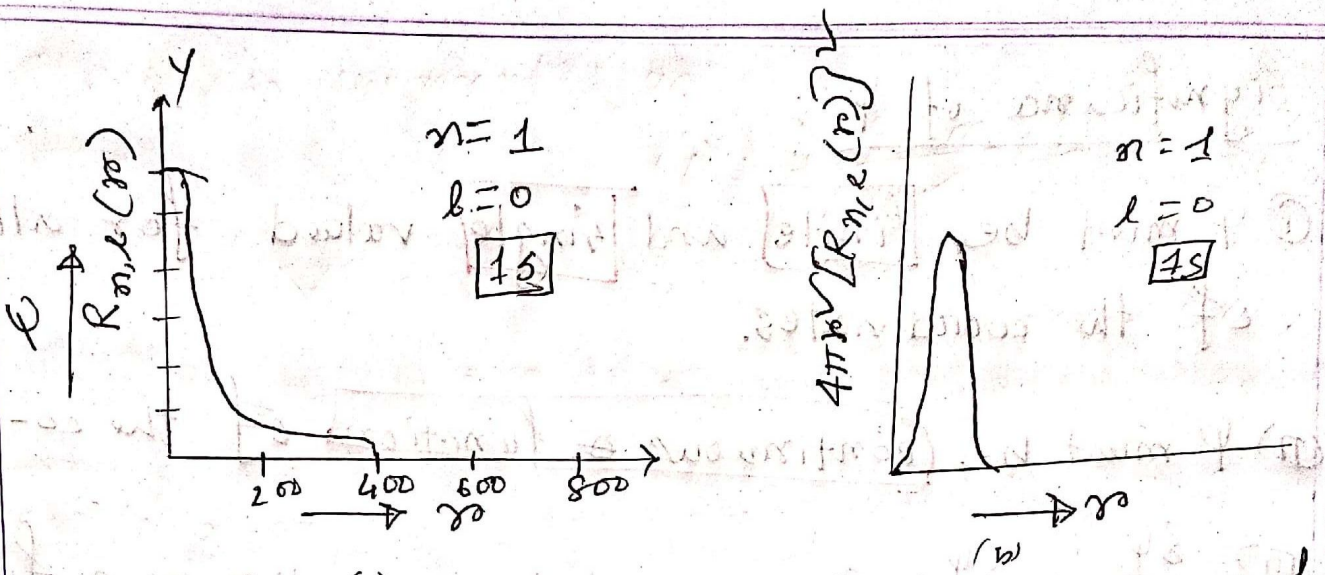


Fig: (a) distribution curve for radial wave function and (b) distribution/probability function.

(c) Why the wave function ψ is squared? Point out the significance of ψ and ψ^2. (2)

Ans:

The wave function ψ is squared because the square of ψ represents the probability of finding a particle at a given location in space, ensuring that the ~~prope~~ probability is always a positive value, unlike the ψ which can be complex and have negative value. This interpretation is known as Born rule.

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Significance of ψ :

(i) ψ must be finite and single valued for all value of the coordinates.

(II) ψ must be continuous function of the co-ordin

(III) $\frac{\delta\psi}{\delta x}$, $\frac{\delta\psi}{\delta y}$, $\frac{\delta\psi}{\delta z}$ must be continuous functio of x, y, z respectively.

(IV)

Significance of ψ^2 :

(i) The probability of finding the e- at any point is proportional to the value of ψ^2 at that point.

(II) The square of the wave function at any point is interpreted as the intensity of the effect at that point. i.e. $I \propto \psi^2$

(III) The value of ψ^2 at any point around the nucleus gives a measure of the electronic charge density at that point.

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(d) (i) Define Angular function $\Theta(\theta)\Phi(\varphi)$ and Radial function $R(r)$. What part of an orbital is signified by these two factors?

(ii) Show that if $\psi = A\sin nx + B\cos 5x$ (where, A, B, n and 5 are constants), then the wave ~~function~~ eqn for a particle in a 1-dimensional box is $\psi = A\sin nx$ ~~$A\sin nx$~~

(i)

In an orbital, the "angular function" signifies the shape of the orbital, describing its ~~interpretation~~ orientation in space and the no. of number of lobes, while the "radial function" indicates the e- distance from the nucleus, essentially defining the size of the orbital and the probability of finding an e- at a certain distance from the nucleus.

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Final 2020

(a) State the zero-point energy. (2)

Ans:

The point in which any particle exists as a most possible lower energy level then that point is called zero point energy.

From the energy expression of a particle of mass "m" moving in a cubic box of length 'a' is given by -

$$E = \frac{h^2}{8ma^2}\left[n_x^2 + n_y^2 + n_z^2\right] \quad \ldots\ldots (i)$$

where n_x, n_y and n_z are the quantum numbers of x, y and z directions respectively.

Now, zero point energy is possible for ground state having quantum numbers 1,1,1 for (n_x, n_y, n_z)

$$\text{So, } E = \frac{h^2}{8ma^2}(1^2 + 1^2 + 1^2)$$

$$= \frac{3h^2}{8ma^2}$$

This energy is called zero point energy.

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(b) What is the physical interpretation of the wave function? Mention the acceptable conditions of ψ. → (4)

* Conditions have discussed.

Then, Physical interpretation of the wave functions:

In quantum mechanics, a wave function represents the state of a particle or system, and its physical interpretation is that the square of its absolute value, $|\psi|^2$, gives the probability density of finding the particle at a specific location in space at a given time; essentially, it describes the likelihood of observing a particle at a certain point, not its exact position due to the wave-particle duality concept.

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(e) Briefly explain the results of the particle in one-dimensional box calculation. (5)

Ans: The Q. gives 5 marks, so the discussion of results is not enough. Hence, derivation / calculation should also be written for final exam which is noted down in Abser sir sheets.

The results of the particle in one dimensional box calculation are of interest with regard to the following two points —

(i) Unlike the classical prediction, the probability of finding the particle is not constant but is a function of x, furtherforce, the probability of finding the particle in a particular portion of the box depends upon the energy of the particle.

(ii) Another difference from classical prediction is the fact that only certain energies, related to the quantum numbers n are allowed. Zero energy is not allowed and the energy increases as n^2.

[Because $n \neq 0$, otherwise $\psi = 0$ and the solution is trivial — the probability of finding the particle is zero ($\psi^2 = 0$) and therefore it does not exist!]

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(n-1)-ℓ

(d) How many radial nodes does each of the following orbitals possess:

(i) 2s (ii) 4s (iii) 3p (iv) 5d → (3)

Ans:

nodes → probability of finding e⁻ is 0

	n	nodes (n-1)	Angular nodes (ℓ)	Radial node
2s	2	1	0	1
4s	4	3	0	3
3p	3	2	1	1
5d	5	4	2	2

Formula: No. of radial nodes = $(n-\ell-1)$

or $(n-1)-\ell$

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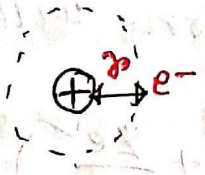
* Final → 2019 *

(a) Write the Schrodinger wave equation for the H atom in Cartesian coordinate system and explain the terms in it. → (3)

Ans:

Schrodinger equation for H atom in cartesian coordinate system:

Bohor atom model of H atom —



The potential energy of the electron of hydrogen atom depends on r — the distance between the nucleus and the electron. To bring an electron from infinity (where the influence of the nucleus charge is zero) to a distance r, the work doone will be—

$$\int_{\infty}^{r_0} F(r_0)\, dr = \int_{\infty}^{r_0} \frac{e^2}{r^2}\, dr_0 = -e^2/r_0$$

And it will be the P.E of the system i.e.

P.E $V = \frac{-e^2}{r_0}$ for hydrogen atom

Now the general wave equation is —

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

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Putting the value,

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}\left(E + \frac{e^2}{r}\right)\psi = 0$$

which is the required equation

For H like systems (e.g. He^+, Li^{2+} etc)
The schrodinger equation will be—

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}\left(E + \frac{Ze^2}{r}\right)\psi = 0$$

Where,

∇ = laplacian operator

$$= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

ψ = Wave function

h = planck's constant

m = mass of particle

E = Total energy $= (T + V)$

Z = Atomic number of atom.

e = charge of e—

r = distance betn e^- and nucleus.

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(a) Explain the terms 'radial distribution function'.

Ans:

→(2)

The radial distribution function (RDF), often denoted as $g(r)$, describes how particles density varies as a function of distance from a reference particle in a system. It provides information about the probability of finding a particle at a certain distance, r from another particle, compared to the probality expected for a completely random distribution.

- Mathematically

$$g(r) = \frac{\rho(r)}{\rho_0}$$

where,

- $\rho(r)$ is the local particle density at a distance r from a reference particle
- ρ_0 is the average particle density of the system.

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(d) Use Slater's rule to estimate values of Z_{eff} for (i) a 4s (ii) a 3d electron in a V atom ($Z = 23$)

Ans:

(i) Effective nuclear charge (Z_{eff}):

The effective nuclear charge is the net positive charge experienced by an electron towards the nucleus in an atom. It is denoted by Z_{eff}.

$$Z_{eff} = Z - \sigma/s$$

where,

Z is the nuclear charge.

σ/s shield or screening constant.

s

$n-1 \longrightarrow 0.85$

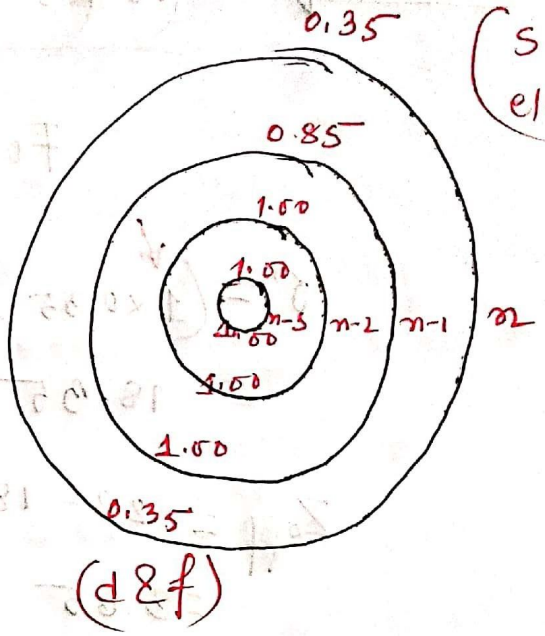
$n \longrightarrow 0.35$

$n-2 \longrightarrow 1.00$

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Trick's for Slater's rule:

e⁻ in orbital	Per electron in orbital		
	n	n-1	n-2 / n-3 etc
s or p	0.35	0.85	1.00
d or f	0.35	1.00	1.00



Li (3) → $1s^2$ $2s^1$ → contributes nothing

$n-1$ n

$\sigma = 2 \times 0.85 +$

$Z_{eff} = 3 - 1.7$

$= 1.3$

→ contibute nothing

B(5) → $1s^2$ $2s^2$ $2p^1$ Z_{eff} for p

$\sigma = 2 \times 0.85 + 2 \times 0.35 = 2.4$

$Z_{eff} = 5 - 2.4$

$= 2.6$

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Ti (22) → $(1s^2)(2s^2\ 2p^6)(3s^2\ 3p^6)\ 3d^2\ 4s^2$

For 3d

$\sigma = (1 \times 0.35 + 18 \times 1.00)$

$= 18.35$

$Z_{eff} = 22 - 18.35$

$= 3.65$

For V (23) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$

(i) For 4s →

$\sigma = 1 \times 0.35 + 18 \times 0.85 + 3 \times 1.00$

$= 18.65$

$Z_{eff} = 23 - 18.65$

$= 4.35$

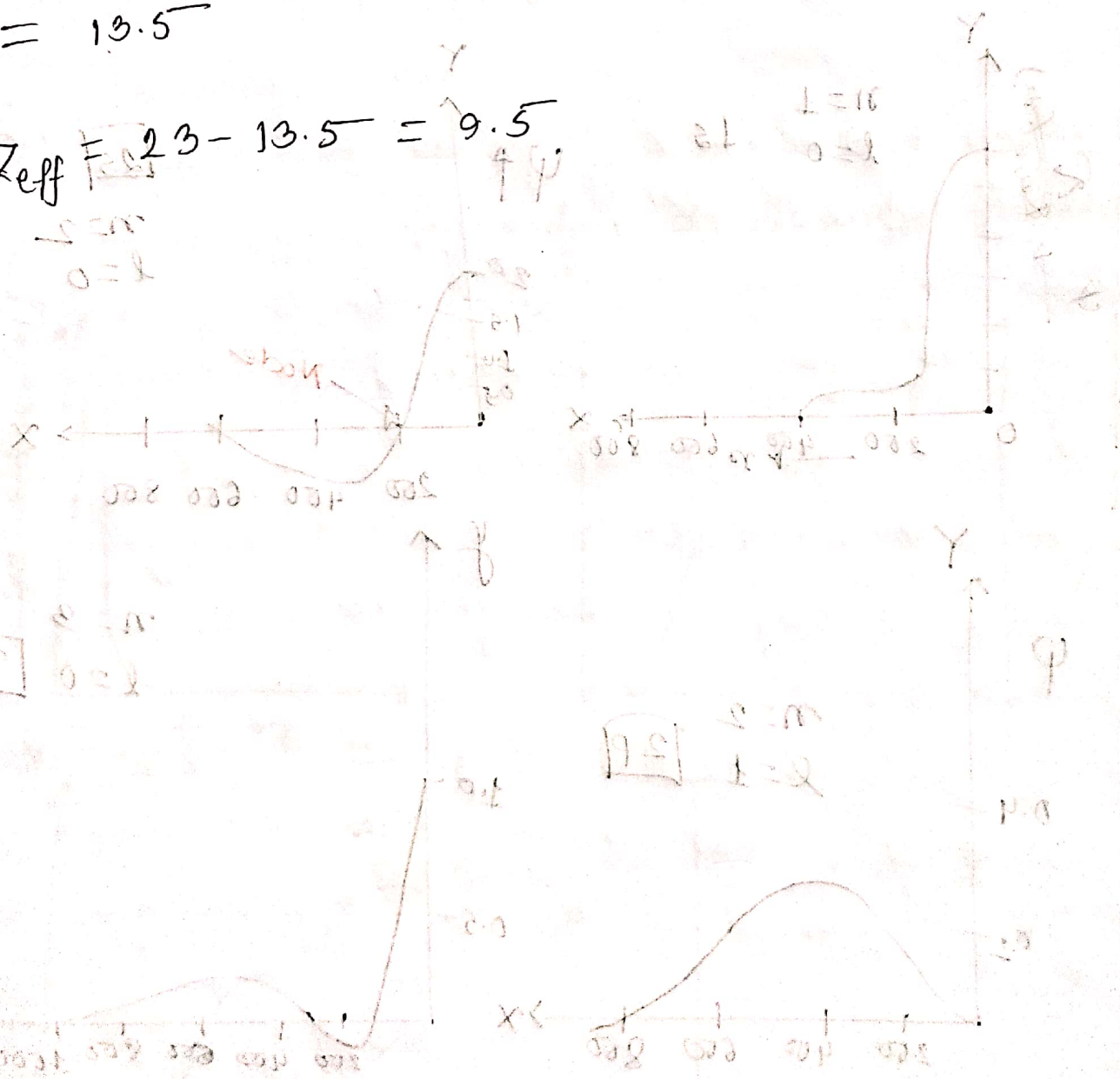
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For 3d → $(1s^2)(2s^2 2p^6)(3s^2 3p^6)3d^2$ ~~$4s^2$~~

$$\sigma = (2\times1 + 8\times1.00 + 8\times0.35 + 2\times0.35)$$

$$= 13.5$$

$$Z_{eff} = 23 - 13.5 = 9.5$$

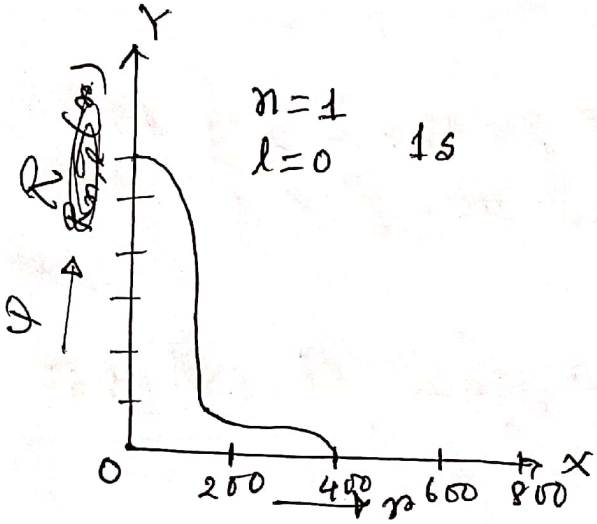


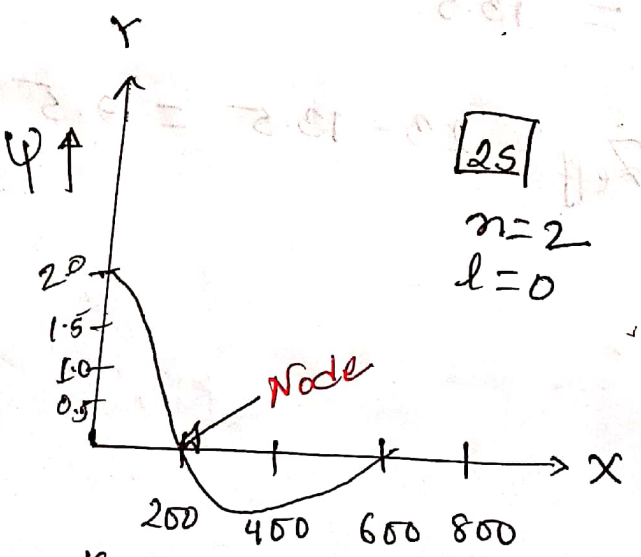
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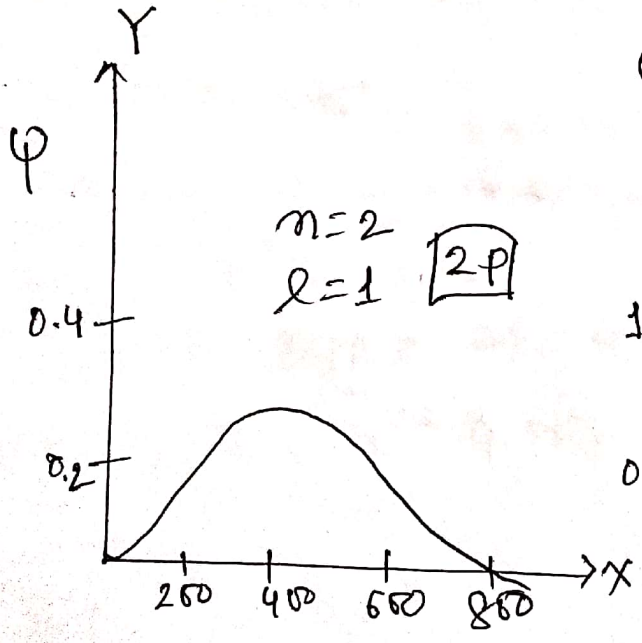
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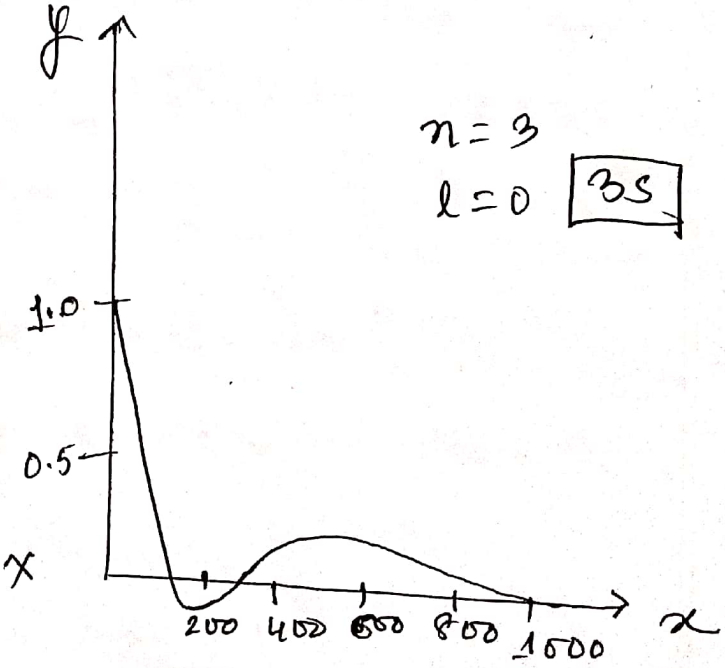
(c) Summarize and represent in figures: (i) radial part of a wave function (ii) radial probability distribution curves of the electron wave function for different electronic states of H atoms.

Distribution curve for radial wave function:

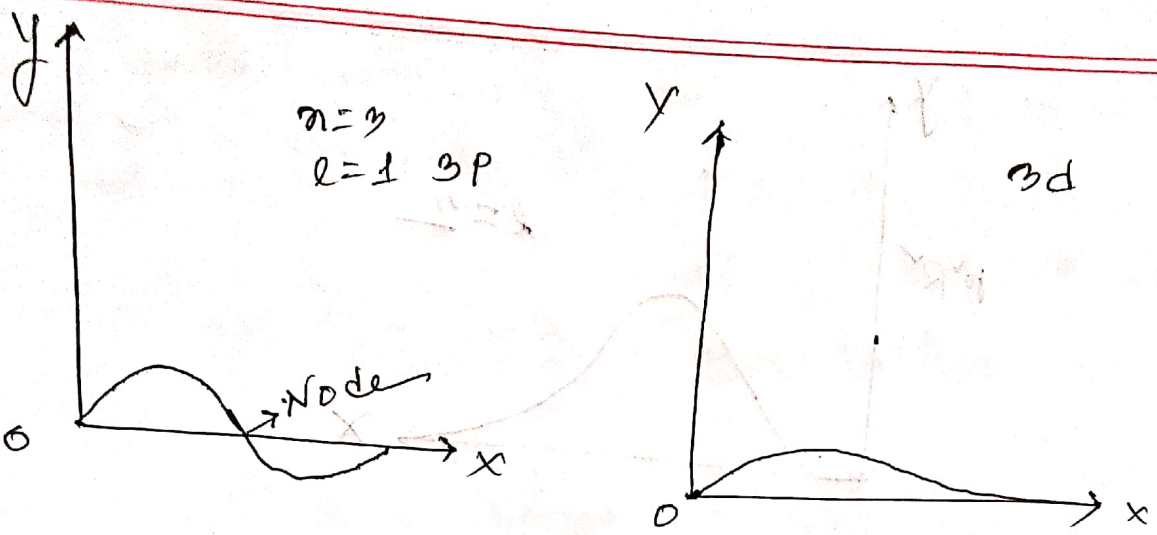




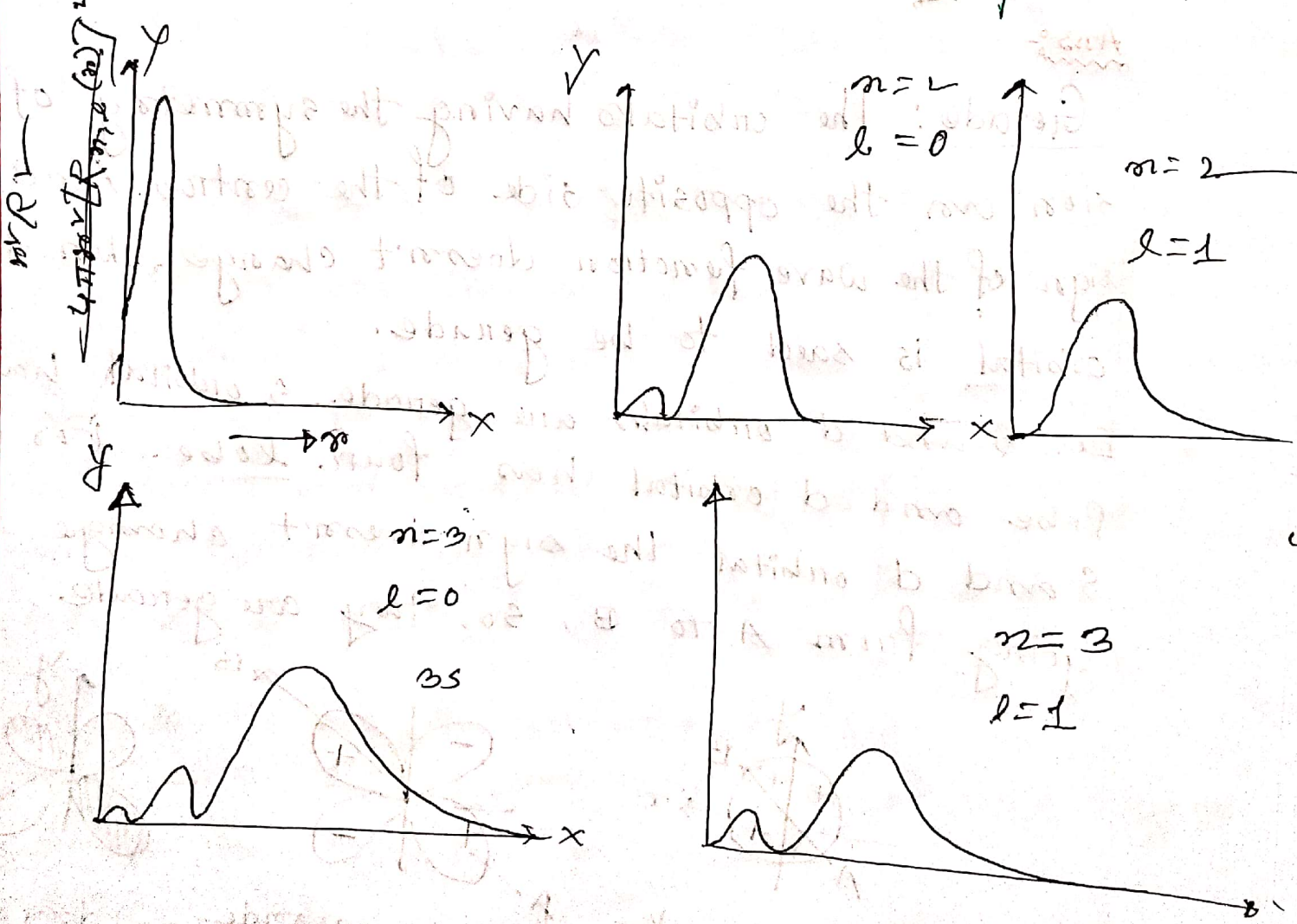




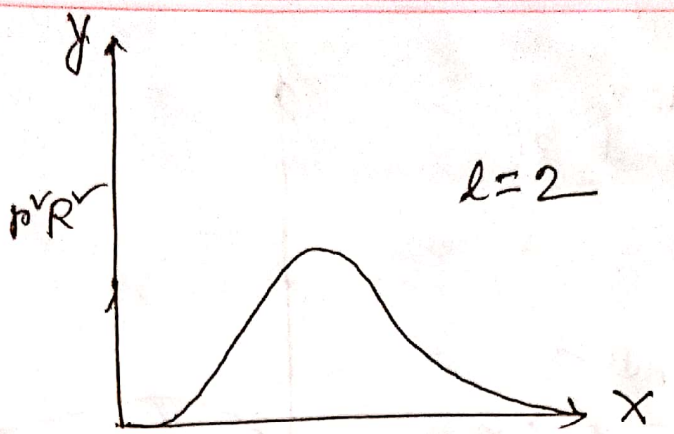
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Distribution Curve for radial distribution function;



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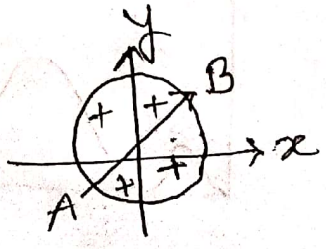


(d) What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples.

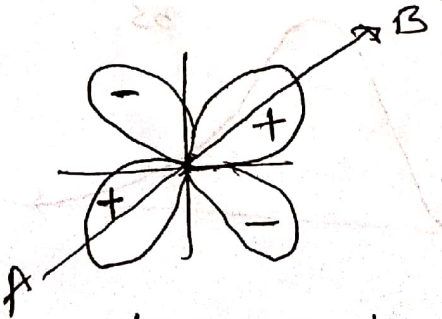
Ans:-

Gerade: The orbitals having the symmetry of inversion on the opposite side of the centre. And if the sign of the wave function doesn't change, then the orbital is said to be gerade.

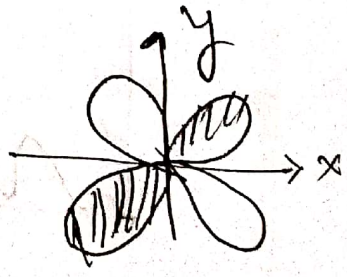
Ex: S and d orbitals are gerade. S orbital has one lobe and d orbital has four lobe. For both S and d orbital the sign doesn't change on going from A to B. So, they are gerade.



S-gerade



dxy-gerade



dxy orbital

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Ungerade: The unsymmetrical orbitals with respect ~~respect~~ to inversion and if the sign of wave func changes then the orbital is said to be ungerade

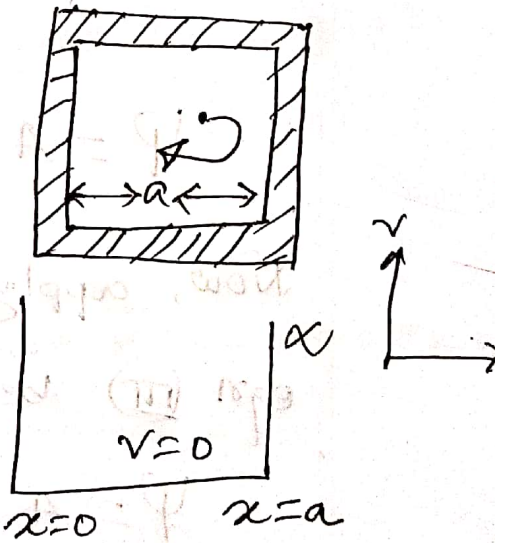
Ex: Final page of this note.

(a) A particle is confined to a one-dimensional box of mass m and length a having infinitely potential wal and its total energy remains constant. Evaluate the energy values E_n of the ~~gerade~~ particle and illustra the energy levels as well as probability distributio functions of the above mentioned box.

Ans: → আবছার স্যারের নোটে বেশ ভালভাবে আছে পরিষ্কার দেখুন,

Particle in one dimensional box:

Let us consider a particle of mass m moving in a straight line along the x-axis and potential energy V inside the box between $x=0$ and $x=a$ is defined as zero



Outside the box, the potential energy is infinite. This means that, the probability of finding the particle outside must be zero when $0 > x > a$

To avoid discontinuity at $x=0$ and $x=a$, the wave

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function must be zero and the boundary conditions be satisfied $\psi = 0$ at $x = 0$ and $\psi = 0$ at $x = a$.

Now, the schrodinger wave eqn for one dimen

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0 \quad \cdots\cdots (1)$$

Since, inside the box potential energy $V = 0$

then, (1) $\Rightarrow$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{8\pi^2 m}{h^2}(E\psi) = 0 \quad \cdots\cdots (II)$$

The solution of this eqn is $\longrightarrow$

$$\psi = A \sin\left(\frac{8\pi^2 mE}{h^2}\right)^{1/2} x + B \cos\left(\frac{8\pi^2 mE}{h^2}\right)^{1/2} x \quad \cdots\cdots (III)$$

Now, applying the condition, $\psi = 0$ at $x = 0$

eqn (III) be comes $\longrightarrow$

$$\psi = A \sin\left(\frac{8\pi^2 mE}{h^2}\right)^{1/2} x \quad \cdots\cdots (IV)$$

Again, applying

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2016

* What is the significance of Schrodinger eqn?

* The wave function described by schrodinger eqn contain all possible information about the quantum system.

* Solving the eqn can determine properties like the probability of finding a particle in a certain location.

(b) What is meant by (i) Well behaved function (ii) Normalization of wave function? → (3)

Ans:

Well behaved function:

A function full filling the following condition is known as well behaved function.

(i) ψ must be finite for all values of x

(ii) ψ must be single valued, that is for each value of x, y must be only one value.

(iii) ψ must be continuous.

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(ii) Normalization of wave function:

If the probability of wave finding the particle in overall space is equal to one, then the funct is called normalized wave function and the process is called normalization of a wave function.

Thus,

$$\int_{x=0}^{x=a} \psi_n^2 \, dx = 1 \quad \ldots\ldots (1)$$

The schrodinger eqn for a particle in one dimensional box,

$$\psi_n = A \sin\left(\frac{n\pi}{a}\right)x \quad \ldots\ldots (II)$$

Putting this value in eqn (1) we get,

$$\int_0^a A^2 \sin^2\left(\frac{n\pi}{a}\right)x \, dx = 1$$

$$\Rightarrow \frac{1}{2} A^2 \int_0^a 2\sin^2\left(\frac{n\pi}{a}\right)x \, dx = 1$$

$$\Rightarrow \frac{1}{2} A^2 \int_0^a \left[1 - \cos\left(\frac{2n\pi}{a}\right)x\right] dx = 1$$

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$$\frac{A^2}{2}\left[\int_0^a 1\,dx - \int_0^a \cos\left(\frac{2n\pi}{a}x\right)dx\right] = 1$$

$$\Rightarrow \frac{A^2}{2}\left\{[x]_0^a + \left[\frac{\sin\frac{2n\pi}{a}x}{\frac{2n\pi}{a}}\right]_0^a\right\} = 1$$

$$\Rightarrow \frac{A^2}{2}(a-0) = 1$$

$$\Rightarrow A^2 = 2/a$$

$$\therefore A = \sqrt{2/a}$$

Now,

$$\psi_n = \frac{\sqrt{2}}{\sqrt{a}}\sin\left(\frac{n\pi}{a}\right)x$$

where, ψ_n is called normalized wave function.

(c) Write the Schrodinger equation for a particle in one-D box and of 3-D box. → 2

Ans:

Schrodinger eqn for a particle in one D.

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

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(d) Gen. Schrodinger eqn for 3-D.

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2 m}{h^2}(E-V)\psi=0$$

*(d) Given the energy expression for a-particle of mass 'm' moving in a cubic box of length 'a'; Hence (i) calculate the zeropoint energy (ii) show that 2nd energy level is degenerate.

Ans:

From the energy expression of a particle of mass 'm' moving in a cubic box of length 'a' is given by —

$$E=\frac{h^2}{8ma^2}(n_x^2+n_y^2+n_z^2)$$

where, n_x, n_y and n_z are the quantum numbers of x, y, z directions respectively.

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Now for (b)

(i) The zero point energy is possible for ground state having quantum numbers 1,1,1 for (n_x, n_y, n_z) respectively.

So, $$E = \frac{h^2}{8ma^2}(1^2+1^2+1^2)$$

$$= \frac{3h^2}{8ma^2}$$

This is the zero point energy.

(ii) Second energy level is (triply) degenerate.

For 2nd energy level, having a possible set of quantum numbers $\psi(2,1,1)$ for (n_x, n_y, n_z), the value of energy will be $\frac{6h^2}{8ma^2}$

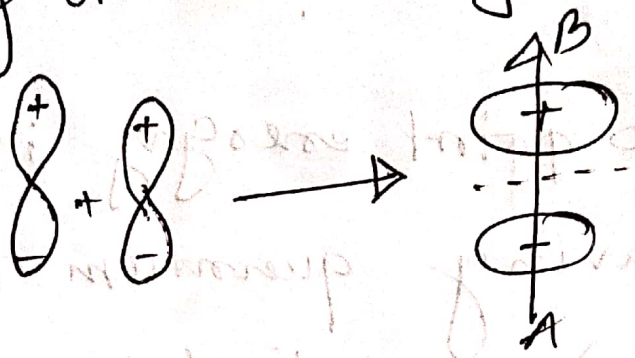
similarly, for another two independent states having quantum numbers $\psi(1,2,1)$ and $\psi(1,1,2)$ for (n_x, n_y, n_z) the energy value will be $\frac{6h^2}{8ma^2}$.

SUZON-50

THE END

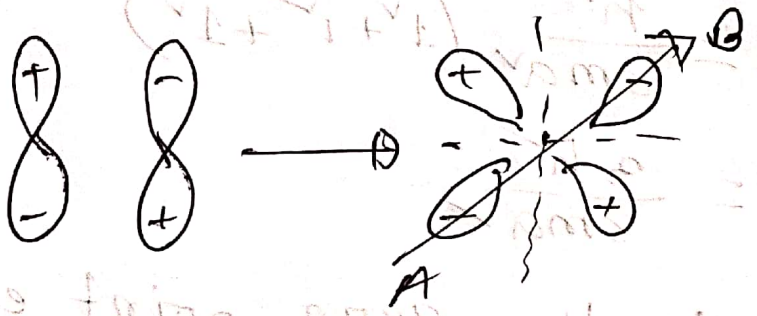
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π bonding orbitals are ~~g~~ ungerade



Ungerade.

π Antibonding orbitals are gerade.



Gerade

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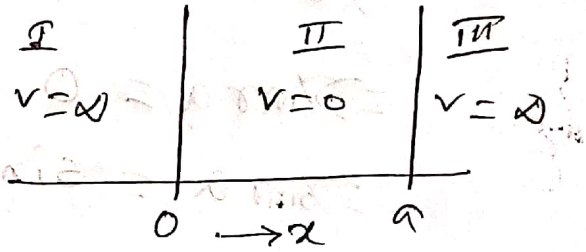
(II)

Given that, SUZON-50

$$\psi = A\sin rx + B\cos sx \quad \text{---} \quad (1)$$

where, A, B, r, s are constant

Verification: (যাচাইকরণ)



$\because \psi = A\sin rx + B\cos sx$

$\therefore \frac{\partial\psi}{\partial x} = Ar\cos rx - Bs\sin sx$

$\Rightarrow \frac{d^2\psi}{\partial^2 x^2} = -Ar^2\sin rx - Bs^2\cos sx$

$= -(Ar^2\sin rx + Bs^2\cos sx)$

Not verified, if $r=s$ then, $\frac{d^2\psi}{\partial^2 x^2} + r^2\psi = 0$

From the boundary conditions it is equivalent that

$\psi = 0$ at $x = 0$ and also at $x = a$

Using the boundary conditions, from eq. (1), we get –

(i) When $x = 0$

$0 = A\sin r.0 + B\cos s.0$

$\Rightarrow B\cos 0^\circ = 0$

$\Rightarrow B = 0$

$\therefore \psi = A\sin rx$ ---- (2)

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(II) When $x = a \neq 0$ from eq (2) we get

$$0 = A \sin \gamma x$$

$$\Rightarrow 0 = A \sin \gamma a$$

$$\Rightarrow \sin \gamma a = 0$$

$$\Rightarrow \sin \gamma a = \sin n\pi$$

$$\Rightarrow \gamma a = n\pi$$

$$\therefore \gamma = \frac{n\pi}{a} \qquad n = 1, 2, 3, \ldots$$

and can not be zero ($n \neq 0$)

$A \neq 0$ because if $A = 0$ then $\varphi = 0$ for all values of a which is not true

$\therefore$ Then eq (2) gives.

$$\varphi = A \sin \frac{n\pi x}{a} \qquad \ldots (3)$$

$$n = 1, 2, 3 \ldots$$

$$\therefore \varphi = A \sin \gamma x$$

(Showed)

Final → 2021 → Same ans 2022.

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